

Should This SBA Loan Be Approved?

A profit-based approach to default prediction

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VinUniversity | Machine Learning for Business

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866,082 loans | \$1.096B validation profit | 77.8% approval at the final cutoff

The Lender's Decision

A bad approval costs five times what a good approval pays.

Decision	Borrower pays	Borrower defaults
Approve	+5% of loan amount	-25% of loan amount
Deny	\$0	\$0

One default cancels the profit from five good loans of equal size.

Profit Objective

Set expected profit on a single approval to zero and solve for the break-even default probability:

$$0.05(1 - p^*) - 0.25p^* = 0 \Rightarrow p^* = \frac{1}{6} \approx 16.67\%$$

- Below 16.67% predicted PD, approving has positive expected value.
- Above it, approving loses money.
- Accuracy treats both errors as equal, while the payoff table makes default errors five times larger.
- Model choice and cutoff choice both follow validation profit.

The Dataset

SBA 7(a) loans, 1987–2014

Public loan-level data from the U.S. Small Business Administration. Target: paid in full vs. charged off.

Loans	866,082
Feature blocks	37 numeric + 7 categorical
Linear / MLP matrix	183 one-hot columns
Overall default rate	17.27% (training)
Pre-2008	14.3% DR
2008–2009 (crisis)	29.3% DR
Post-2009	8.7% DR

Two layers of features

Loan	Term, gross approval, guaranteed share, RE flag
Borrower	Employees, business age, jobs created/retained
Geography	Borrower state, lender state, urban/rural
Industry	NAICS sector
Program	LowDoc, revolving credit
External	<i>HPI, CPI, oil, income, bank failures, FEMA, QCEW</i>

External variables are merged using only information observable on or before each loan's approval date.

Data Preprocessing Pipeline

A clean pipeline separates three roles: application signals for prediction, external context at approval time, and loan amount for profit weighting.

Step	What happens	Final view	Count
1. Start	866,082 SBA approvals with 27 base columns.	Trees	44 features
2. Context	HPI, CPI, oil, income, bank failures, FEMA, and QCEW are merged by approval date, state, and sector.	Linear / MLP	183 one-hot columns
3. Features	Outcome fields, charge-off fields, and redundant raw columns are removed; 13 interaction features are engineered.	Profit	DisbursementGross
4. Split	Stratified $y \times \text{era}$ split: 554K train, 139K val, 173K test.	Target	Paid vs charged off

The split and profit-weighting logic stay fixed across model families, so the leaderboard compares model behavior on the same prepared dataset.

Feature Engineering: Application Variables

Most signal is already in the application. Feature engineering makes that signal cleaner without inventing post-approval information.

Feature step	Variables created	Why it is used
Log scale	log gross approval; log employees	Compresses extreme loan-size and firm-size tails without changing rank order.
Structural ratios	SBA guarantee share; jobs per dollar	Preserves application-time loan structure and cuts raw duplicate amount fields.
Approval-time flags	real-estate-like long maturity; same-state lender; recent bank failure	Turns recurring EDA patterns into explicit model signals.

Trees use raw numeric scale; linear/MLP models use standardized numeric features and one-hot categories. Disbursement amount is excluded from predictors and used only as the dollar exposure in profit.

External Features and Interactions

Outside data is added only when it matches a story from EDA and is known by the approval date.

External source	Approval-time feature	Interaction tested
HPI	falling prices; 1-year growth; 2-year draw-down	construction and real-estate-like loans under housing stress
CPI / Oil	inflation and oil-price shock flags	micro firms and supply-chain sectors under cost pressure
Income	1-year income growth; income-decline flag	consumer-facing sectors under weak demand
FEMA	hurricane, fire, flood, storm counts	tourism, coastal, and disaster-sensitive sectors after shocks
FDIC / QCEW	recent bank failures; industry contraction	local credit tightening and weak state-industry conditions

VIF and Duplication Screen

Multicollinearity cleanup belongs before model training. Robustness checks then focus on the final approval rule.

VIF / duplication screen

Reason moved	re-	Variables removed
Raw amount duplicates		GrAppv, SBA Appv, NoEmp, log SBA Appv
Algebraic duplicates	dupli-	unguaranteed ratio, unguaranteed amount, jobs total
Raw calendar fields		approval year, ApprovalFY clean
Redundant win-dow	win-	state bank failures 24m

Clean modeling rule

- Raw year fields are removed from predictors.
- Outcome and charge-off fields are removed from predictors.
- Disbursement amount is reserved for profit exposure.
- Final matrix is 37 numeric + 7 categorical features.

Leakage Control

Before modeling, every post-approval and outcome-dependent column is excluded from the feature set.

Variable	Decision	Reason
MIS_Status	Target only	Outcome label; never a feature
ChgOffDate	Drop	Recorded only after default
ChgOffPrinGr	Drop	Charged-off amount; conditional on outcome
BalanceGross	Drop	Outstanding balance at settlement
DisbursementGross	Drop as feature	Set after approval; retained only as dollar weight in profit

Splits

Stratified by outcome $\times$ era. Test sealed until final evaluation.

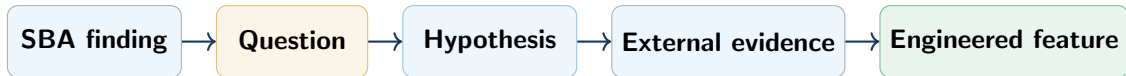
Train	554,292
Validation	138,573
Held-out test	173,217

External merge rule

Each loan sees the 12 months *before* approval. A March 2008 approval can use data through February 2008.

Three Threads, One Method

The EDA follows three threads: start with a pattern in the SBA file, ask what the row is missing, then check whether an external source explains the gap.

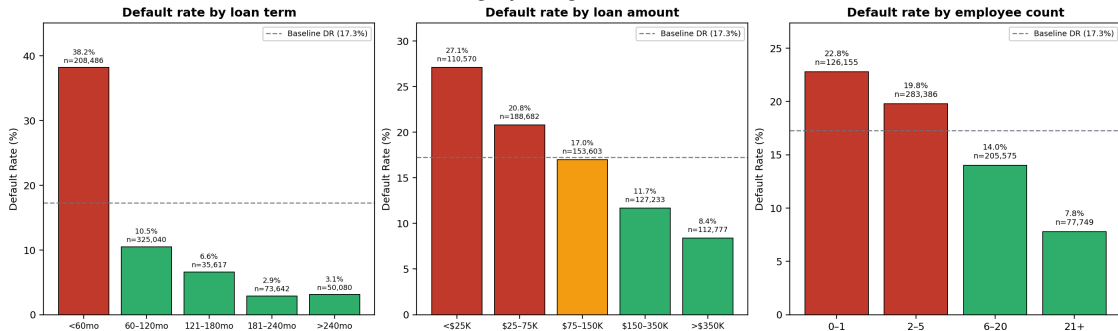


1. Borrower fragility under macro stress
2. Industry cyclicalities conditional on local stress
3. Geographic exposure conditional on disaster type

Original SBA Clues: Fragile Borrowers Stand Out Early

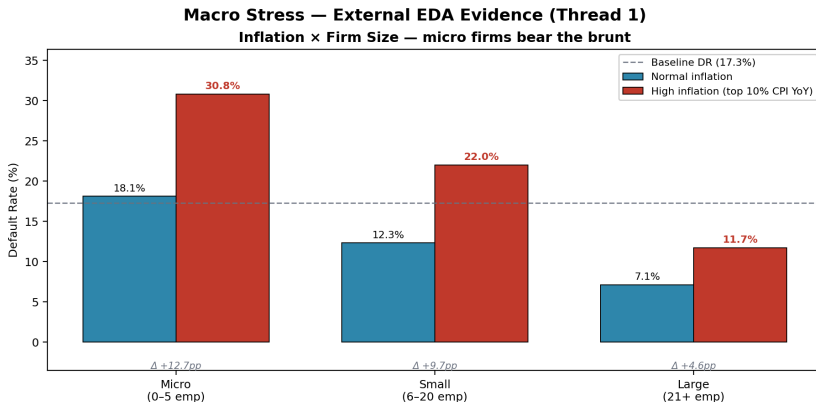
*The SBA file already points to a fragile-borrower story: short maturities, small balances, and micro firms default far above baseline. **Question:** does macro stress widen that fragility gap?*

Borrower Fragility — Original SBA Risk Patterns



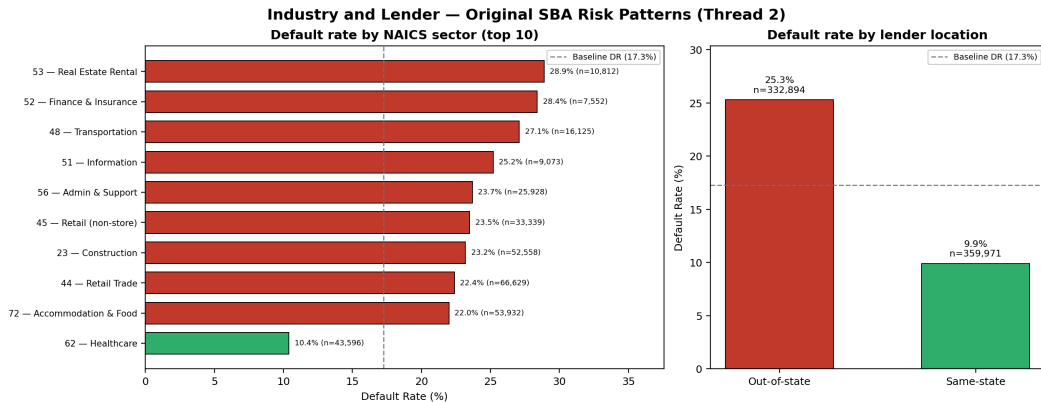
External EDA 1: Macro Stress Sharpens Borrower Fragility

Inflation should hit thin-margin borrowers first. Micro firms have less room to absorb cost shocks, so their default rate should move earlier.



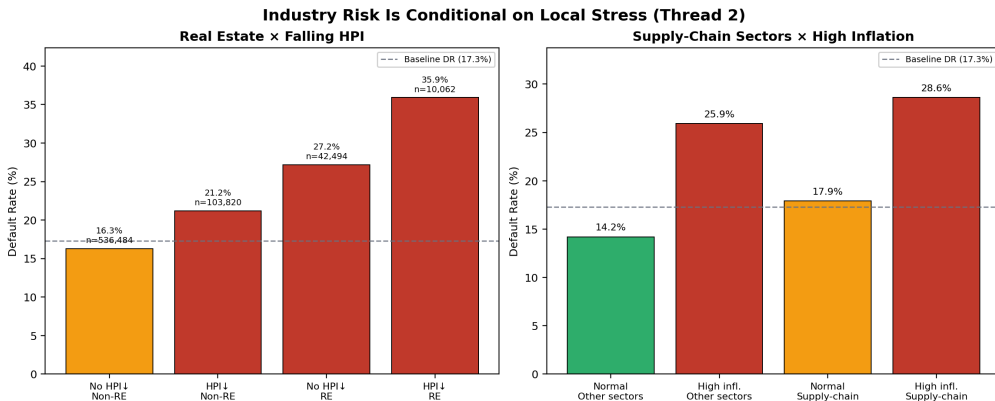
Industry and Lender Setup Narrow the Search

Cyclical industries and out-of-state lenders cluster at the risky end. Question: is that risk permanent, or does it spike when local conditions weaken?



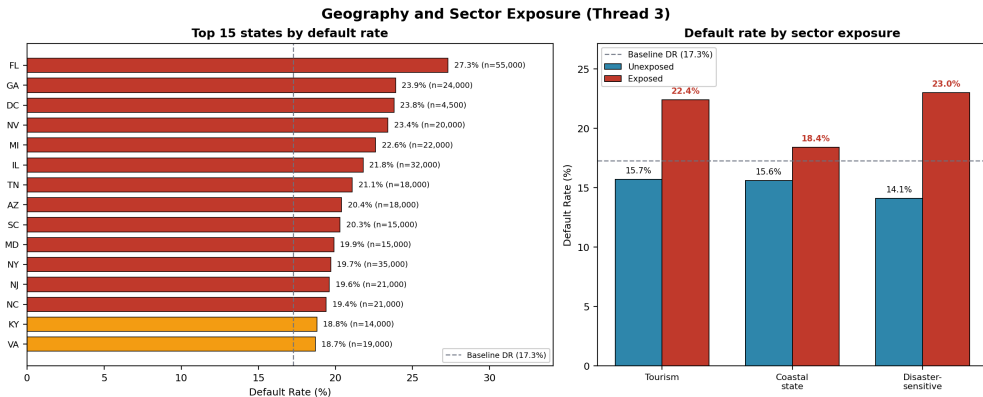
External EDA 2: Housing Stress Creates Worse Industry Corners

Industry labels become more useful once the local cycle enters the picture: construction and real estate under falling HPI, supply-chain sectors under inflation.



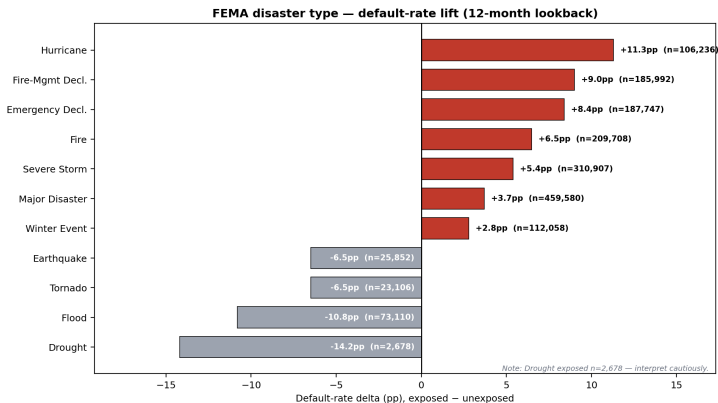
Geography Already Looks Like Exposure

Even before FEMA, the geography has structure: Sun Belt, coastal, tourism, and disaster-sensitive sectors sit above baseline. **Question:** *which shocks actually explain that exposure?*



Disaster Type Matters More Than a Generic Flag

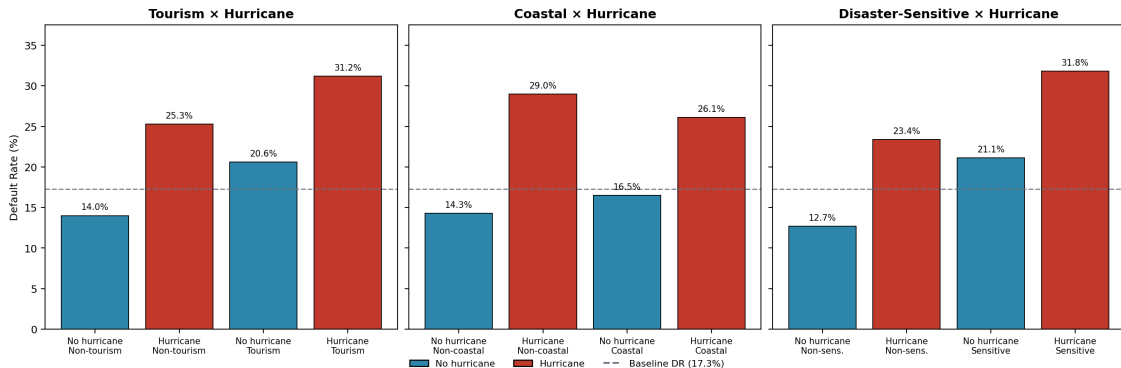
The FEMA merge separates damaging shocks from noisy labels. Hurricanes, fire, and severe storms raise default; flood and drought behave differently.



Hurricane Risk Depends on Who Gets Hit

The cleanest disaster signal appears after conditioning on who is exposed: tourism, coastal, and disaster-sensitive sectors deteriorate most after hurricanes.

Hurricane Interactions — Thread 3 external



From Three Threads to Thirteen Engineered Features

We kept the interactions that the charts could defend and left the rest for reporting.

Channel	SBA clue	External hypothesis	Engineered feature(s)
Borrower fragility	Micro firms, short maturities default 2–10× baseline	Macro pressure hurts thin-margin borrowers	<code>inflation_micro_business</code>
Supply stress	Cyclical sectors at risky end	Oil/cost shocks bite supply chains	<code>oil_shock_supply_chain</code> , <code>supply_chain_inflation</code>
Local demand	Consumer sectors elevated	Real-income decline cuts discretionary demand	<code>income_decline_consumer_sector</code>
Credit supply	Geographic risk concentration	Bank failures signal local tightening	<code>recent_bank_failure_flag</code>
Housing stress	Construction & RE default +20–35% in falling-HPI	Falling HPI weakens collateral and demand	<code>construction_hpi</code> , <code>real_estate_industry_hpi</code>
Geo exposure	Sun Belt, coastal states elevated	Geography captures disaster + labor exposure	<code>tourism_industry_flag</code> , <code>coastal_state_flag</code> , <code>disaster_sensitive_sector_flag</code>
Disaster × sector	Hurricane flood/drought (selection) +11pp; reversed	Hurricanes destroy capital; sector decides recovery	<code>tourism_hurricane</code> , <code>coastal_hurricane</code> , <code>disaster_sensitive_hurricane</code>

Report-only features: generic disaster flag, raw stress index, flood / drought / earthquake (selection effects).

13 interactions + 31 base = 44 features → **183 columns** after one-hot.

Modeling Approach

To keep the comparison honest, interpretable baselines, tree ensembles, and a neural benchmark all face one split and one profit rule.

Family	Reason to include	Expected weakness
Logistic, LDA, QDA	Interpretable baseline; regulatory standard	One global linear effect per variable
RF, bagging, HGB, XG-Boost, LightGBM	Capture thresholds and interactions on tabular data	Less interpretable; sensitive to imbalance
Multi-layer perceptron	Nonlinear benchmark beyond trees	Boosters usually win on tabular features

Reported metrics cover both ranking quality and the lending decision: profit, AUC, Brier score, approval rate, and approved default rate.

Statistical Baselines

Models

Logistic regression (Ridge, Lasso, ElasticNet); LDA with shrinkage; QDA with covariance regularization.

Ridge solver choice

Quasi-Newton method – approximates the inverse Hessian from recent gradient updates without ever building the 183×183 matrix. Stable when features have mixed scales (standardized continuous next to binary one-hots). Converges in ~ 25 iterations; plain SGD would be noisier and need careful LR scheduling.

Sparse-penalty solver choice

L1 is non-smooth at zero – vanilla gradient methods break at the corner. SAGA stochastically averages past gradients; it's the standard L1 solver.

Best validation results

Model	Profit	AUC
Ridge ($C=10$)	\$791.3M	0.875
Ridge ($C=1$)	\$791.4M	0.875
Ridge ($C=0.1$)	\$790.8M	0.875
LDA	\$769.3M	0.861

Limitation: one global linear effect per variable can't represent interaction-heavy risk patterns. The \$287M gap to LightGBM measures how much nonlinearity matters here.

Tree-Based and Boosting Models

Two-stage tuning

Stage 1 – manual grid. 14 configs across RF/HGB/XGB/LGB chosen by data-size intuition: $n_{\text{est}} \in \{100, 300\}$, depth/leaves wide enough for interactions, `min_samples_leaf` large enough to suppress noise splits.

Stage 2 – Optuna TPE. Splits trials into top 25% and bottom 75%, fits densities $\ell(\theta)$ and $g(\theta)$ over each, then samples next trial maximizing $\ell(\theta)/g(\theta)$ – searches where good is dense and bad is sparse.

Search budget. Thirty trials keep the rerun affordable while still allowing late improvements; the best trial arrives near the end of the search.

Solver. Functional gradient descent – trees fit the loss gradient (XGBoost adds Hessian for splits).

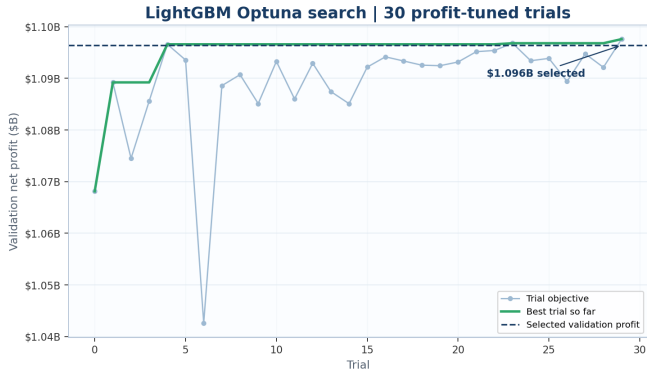
Top results on validation

Model	Profit	Time
LGB n300 nl127	\$1,078.6M	12s
LGB n300 nl63	\$1,072.3M	6s
Bagging n100	\$1,062.3M	54s
XGB n300 d10	\$1,058.1M	6s
LGB n200 nl31	\$1,055.1M	4s
HGB i300 d63	\$1,044.3M	27s
AdaBoost n100	\$1,020.0M	132s

Manual benchmark winner: `n_est=300`, `num_leaves=127`, `lr=0.05`, `min_child=50`, `subsample=0.9`. The final Optuna LightGBM improves this to \$1,096.4M with a 418-tree / 194-leaf configuration. RF ranks #13 at \$947.2M.

Final Model: Optuna LightGBM

The benchmark board chose the model family. The final recommendation uses the profit-tuned Optuna LightGBM.



Item	Value
Trials	30
Best trial	29
Validation profit	\$1,096.4M
Validation AUC	0.9827
Brier score	0.0335
PD cutoff	0.1488
Approval rate	77.8%
Approved DR	1.09%
ROI	4.70%
Manual benchmark	\$1,078.6M
Gain vs manual	+\$17.8M

Best params: 418 trees, 194 leaves, learning rate 0.0637, min child 85, subsample 0.502, colsample 1e/38

Neural Network Benchmark

Architecture

MLP 256-128-64-32, ReLU + He init. L2 decay + early stopping (15% holdout); converges by epoch ~ 80 .

Optimizer choice

Features have wildly different gradient magnitudes – a binary LowDoc flag and continuous $\log(\text{GrAppv})$ differ by orders of magnitude. Plain SGD's single LR moves too fast on some weights, too slow on others.

Adam tracks per-parameter m_t (momentum) and v_t (squared gradient), updates $w \leftarrow w - \eta \cdot m_t / (\sqrt{v_t} + \epsilon)$. The $\sqrt{v_t}$ division scales step size per parameter automatically. Bias correction $\div (1 - \beta^t)$ fixes the zero-init downward bias on early steps.

Result on validation

Architecture	Profit	AUC
MLP 256-128-64-32	\$978.5M	0.9595
MLP 128-64-32	\$970.9M	0.9580
MLP 256-128-64	\$972.3M	0.9573

Read

MLP beats every linear model by \$180M+ – nonlinear signal is real. But it trails LightGBM by \$100M because tabular data has sharp decision boundaries (Term ≥ 240 mo flips risk) that trees capture in one split and MLPs need many parameters to approximate.

Validation Leaderboard

Validation metrics for the final Optuna model and the fixed manual benchmark board.

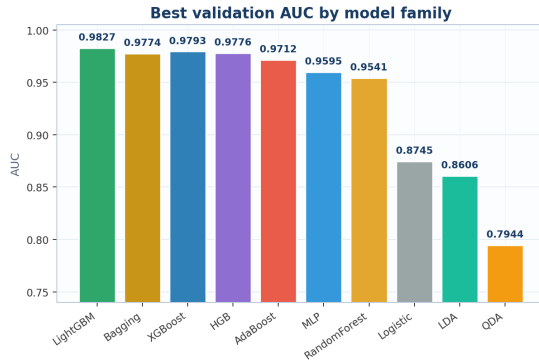
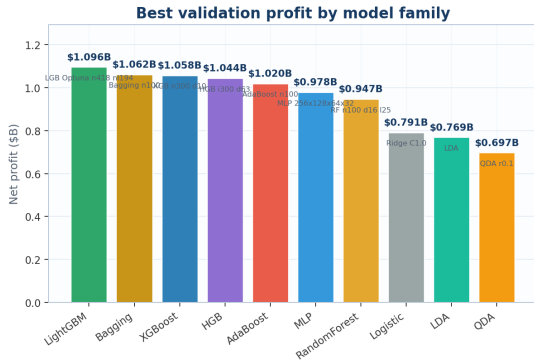
Model	Profit (M\$)	AUC	Brier	Runtime
LGB Optuna n418 nl194	1,096.4	0.9827	0.0335	30 trials
LightGBM n300 nl127	1,078.6	0.9814	0.0351	12s
LightGBM n300 nl63	1,072.3	0.9803	0.0365	6s
Bagging n100	1,062.3	0.9774	0.0399	54s
XGBoost n300 d10	1,058.1	0.9793	0.0367	6s
LightGBM n200 nl31	1,055.1	0.9772	0.0399	4s
HGB i300 d63	1,044.4	0.9776	0.0388	27s
AdaBoost n100	1,020.0	0.9712	0.1561	127s
MLP 256-128-64-32	978.5	0.9595	0.0514	91s
Ridge $C=10$	791.3	0.8745	0.0898	13s
LDA (shrinkage)	769.3	0.8606	0.0959	—

Source: `all_model_results.csv` for the manual board and `optuna_metrics.json` for the final row.

Best Model by Family

Best validation profit and AUC by model family. The LightGBM family includes the final Optuna run.

Best model by family | final Optuna LightGBM included



What Drives the Final Model



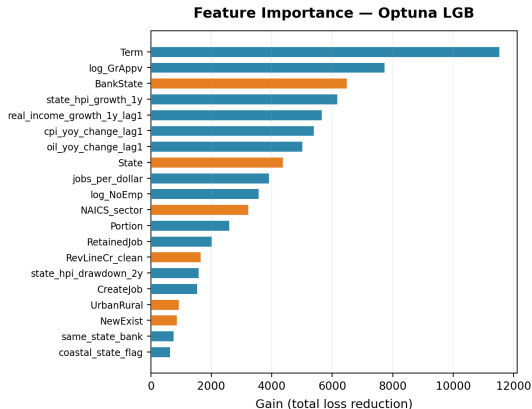
SHAP summary, top 15 features on the validation set.

Reading the chart

- **Term** dominates – the strongest structural risk signal.
- External macro indicators (HPI growth, CPI) carry the temporal signal – no explicit year column is needed.
- **BankState** is the lender's state; **same_state_bank** flags whether borrower and lender are in the same state.
- **UrbanRural** records the SBA urban/rural location code.
- Selected external features survive: HPI growth, recent inflation, oil

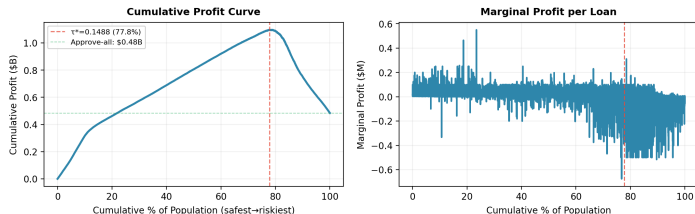
Feature Importance: Same Story from Split Gain

Split gain tells the same story as SHAP, using a different lens: term, lender geography, loan size, HPI, income, state, and macro stress remain near the top.



Profit Curve and the Approval Cutoff

Profit Lift Curve — Optuna LGB (n418 n1194)



Left: cumulative profit as approval depth grows. Right: marginal profit per additional approval.

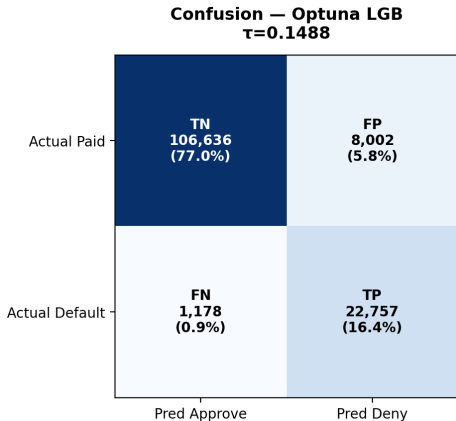
Operating point

Metric	Value
Approve-all baseline	\$483.9M
Optimal profit	\$1,096.4M
Lift from screening	+\$612.5M
PD cutoff τ^*	0.1488
Approval rate	77.8%
Approved default rate	1.09%
ROI	4.70%

Approve until predicted PD exceeds 0.1488. Past that point, the next approvals reduce portfolio profit.

Approval Outcome Matrix

At the selected cutoff, the matrix reads like a lending decision: paid loans approved or lost, defaulted loans approved or avoided.



Validation outcome	Count
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Paid and approved	106,636
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Paid but denied	8,002
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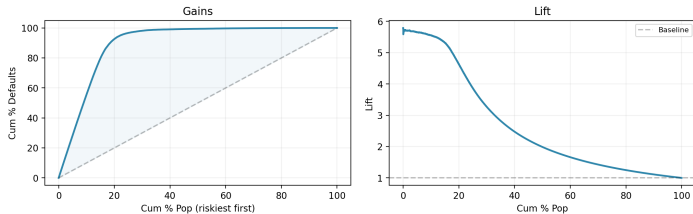
Default but approved	1,178
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Default and denied	22,757
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Approval rate is 77.8%. Approved default rate is 1.09%, with 4.70% ROI. The business move is targeted: keep most of the book, cut the tail where expected profit turns negative.

Gains and Lift

Gains & Lift — Optuna LGB



Left: gains chart. Right: lift chart. Validation set, ordered by predicted PD.

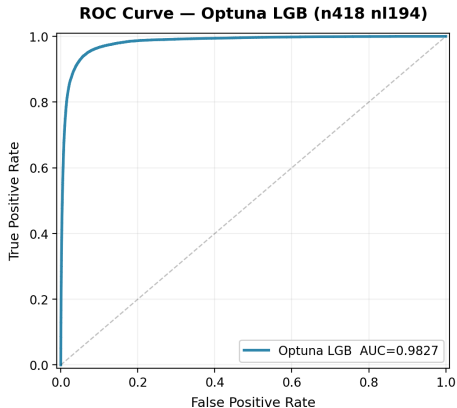
Decile read

Decile	PD range	DR
D1	0.0000–0.0009	0.01%
D5	0.0044–0.0068	0.32%
D7	0.0139–0.0437	1.72%
D8	0.0437–0.2537	10.11%
D9	0.2537–0.8925	63.77%
D10	0.8925–0.9944	95.73%

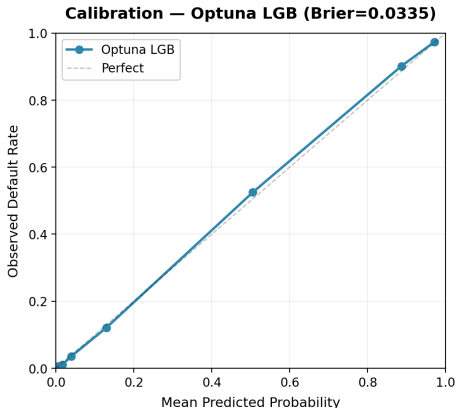
Deciles are equal-sized validation groups sorted from safest (D1) to riskiest (D10). D8 is the last group that still adds to profit.

Ranking and Calibration Check

Profit decides the cutoff. Ranking and calibration checks guard against a bookkeeping artifact.



AUC = 0.9827 on validation.



Brier = 0.0335. Calibration is good enough near the operating cutoff.

Held-Out Test Check

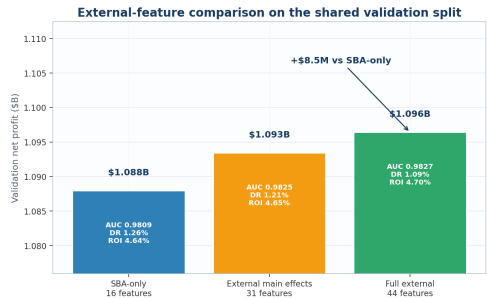
Only after freezing the validation winner and cutoff did we open the 173,217-loan test set.

Metric	Validation	Test
N	138,573	173,217
AUC	0.9827	0.9826
Brier score	0.0335	0.0339
Frozen PD cutoff	0.1488	0.1488
Approval rate	77.8%	77.7%
Approved default rate	1.09%	1.11%
ROI	4.70%	4.67%
Profit	\$1,096.4M	\$1,358.3M

- Test AUC (0.9826) stays essentially equal to validation (0.9827).
- The validation cutoff $\tau^* = 0.1488$ is frozen before test evaluation.
- Approved DR stays near 1.1% on both sets, with ROI near 4.7%.

External Data: Incremental Profit

Same split, same Optuna LightGBM setup, same profit rule.



Validation net profit by feature set.

Feature set	Feat.	Profit	AUC	DR
SBA-only	16	\$1,087.9M	0.9809	1.26%
External main	31	\$1,093.4M	0.9825	1.21%
External full	44	\$1,096.4M	0.9827	1.09%

Feature set	τ	Approval	ROI
SBA-only	0.1687	78.1%	4.64%
External main	0.1637	78.2%	4.65%
External full	0.1488	77.8%	4.70%

Full external specification vs SBA-only: +\$8.5M validation profit.

Robustness Checks

Validation, transport, payoff, seed, feature-ablation, and calibration checks.

Risk ranking and seed checks

Check	N test	DR	AUC
Validation split	138,573	17.3%	0.9827
Held-out test	173,217	17.3%	0.9826
Pre → crisis	50,401	29.3%	0.960
Pre+crisis → post	33,321	8.7%	0.970
Core-signal ablation	–	–	0.838

Seed	Profit	AUC	τ
1	\$1,096.4M	0.9827	0.1488
42	\$1,089.0M	0.9825	0.1637
123	\$1,089.8M	0.9821	0.1885
Mean	\$1,091.7M	–	–

SD = \$4.1M

Payoff and cutoff checks

Loss	τ	Approval	Profit
-15%	0.2331	79.8%	\$1,127.7M
-20%	0.2034	79.2%	\$1,111.2M
-25%	0.1488	77.8%	\$1,096.4M
-30%	0.1488	77.8%	\$1,084.6M
-35%	0.1488	77.8%	\$1,072.7M

Cutoff / calibration metric	Value
Validation cutoff	0.1488
Validation Brier	0.0335
Test profit at frozen cutoff	\$1,358.3M
Test approval rate	77.7%
Test approved DR	1.11%
Platt scaling profit change	<\$1M

Robustness Notes for Q&A

Numerical details behind the robustness checks.

Ranking evidence

Check	AUC
Validation split	0.9827
Held-out test	0.9826
Pre-2008 → crisis	0.960
Pre+crisis → post	0.970
After removing top trio	0.838

Era-forward rows use chronological transport, not the random validation split.

Cutoff evidence

Scenario	τ	Profit
Validation	0.1488	\$1.096B
Held-out test	0.1488	\$1.358B
Manual benchmark	0.1934	\$1.079B
Multi-seed mean	–	\$1.092B ± \$4.1M
Loss –15%	0.2331	\$1.128B
Loss –20%	0.2034	\$1.111B
Loss –30%	0.1488	\$1.085B
Loss –35%	0.1488	\$1.073B

Source: robustness CSVs and PAPER_TRUTH.json.

Appendix: Final Model and Benchmark Board

The first row is the final Optuna model. The remaining rows are the original 21-model manual benchmark board.

Model	Family	Profit M\$	AUC	Brier	τ	App. DR
LGB Optuna n418 nl194	LightGBM	1,096.4	0.9827	0.0335	0.1488	1.09%
LGB n300 nl127	LightGBM	1,078.6	0.9814	0.0351	0.1934	1.39%
LGB n300 nl63	LightGBM	1,072.3	0.9803	0.0365	0.1786	1.35%
LGB n200 nl31	LightGBM	1,055.1	0.9772	0.0399	0.1587	1.32%
RF n200 d16 l50	RF	934.8	0.9520	0.0608	0.1538	2.10%
RF n100 d16 l50	RF	936.0	0.9521	0.0606	0.1538	2.10%
RF n100 d16 l25	RF	947.2	0.9541	0.0592	0.1667	2.23%
XGB n300 d10	XGBoost	1,058.1	0.9793	0.0367	0.1439	1.29%
XGB n200 d8	XGBoost	1,045.5	0.9770	0.0393	0.1637	1.46%
HGB i300 d63	HGB	1,044.4	0.9776	0.0388	0.184	1.61%
HGB i200 d31	HGB	1,032.2	0.9745	0.0420	0.1667	1.57%
Bagging n100	Bagging	1,062.3	0.9774	0.0399	0.1736	1.31%
AdaBoost n100	AdaBoost	1,020.0	0.9712	0.1561	0.4562	1.71%
MLP 256x128x64x32	MLP	978.5	0.9595	0.0514	0.0992	2.24%
MLP 256x128x64	MLP	972.3	0.9573	0.0527	0.1667	2.68%
MLP 128x64x32	MLP	970.9	0.9580	0.0521	0.1240	2.42%
Ridge C10	Logistic	791.3	0.8745	0.0898	0.2232	5.07%
Ridge C1	Logistic	791.4	0.8745	0.0899	0.2232	5.08%
Ridge C0.1	Logistic	790.8	0.8745	0.0899	0.2182	4.99%
LDA	LDA	769.3	0.8606	0.0959	0.208	5.62%
QDA r0.2	QDA	693.0	0.7887	0.1745	0.1042	6.23%
QDA r0.1	QDA	697.4	0.7944	0.1873	0.0893	6.01%

Manual winner: \$1,078.6M at $\tau = 0.1934$. Final profit-tuned Optuna model: \$1,096.4M at $\tau = 0.1488$.

What We Did Not Solve

Approved-only sample. Rejected applicants are absent from the public file. The rule ranks historically approved SBA loans; reject inference remains outside scope.

Random split is easier than deployment. Validation mixes crisis and non-crisis years. A live lender would meet the next shock sequentially; the era-forward test is closer to that setting.

Right-censoring in later vintages. Loans originated near the end of the sample have had less time to fail. That mechanically pushes observed post-crisis default rates down.

Stylized payoff rule. The \$+5% / \$-25% matrix is a classroom decision rule. A production bank P&L would add recovery timing, servicing cost, and loan-specific pricing.

External data are coarse. FEMA is date-specific, while several macro merges are monthly, quarterly, or annual. We use them as approval-time context.

External-feature lift. External features improved profit by \$8.5M on the common split: \$1,087.9M for SBA-only versus \$1,096.4M for the full feature set.

Fairness and deployment remain open. Geography, industry, and disaster exposure may track broader structural disparities. Any real deployment would need subgroup review, manual-review design, and governance before use.

Final Takeaway



Approve 77.8% of loans. Add \$612.5M over approve-all on validation.